



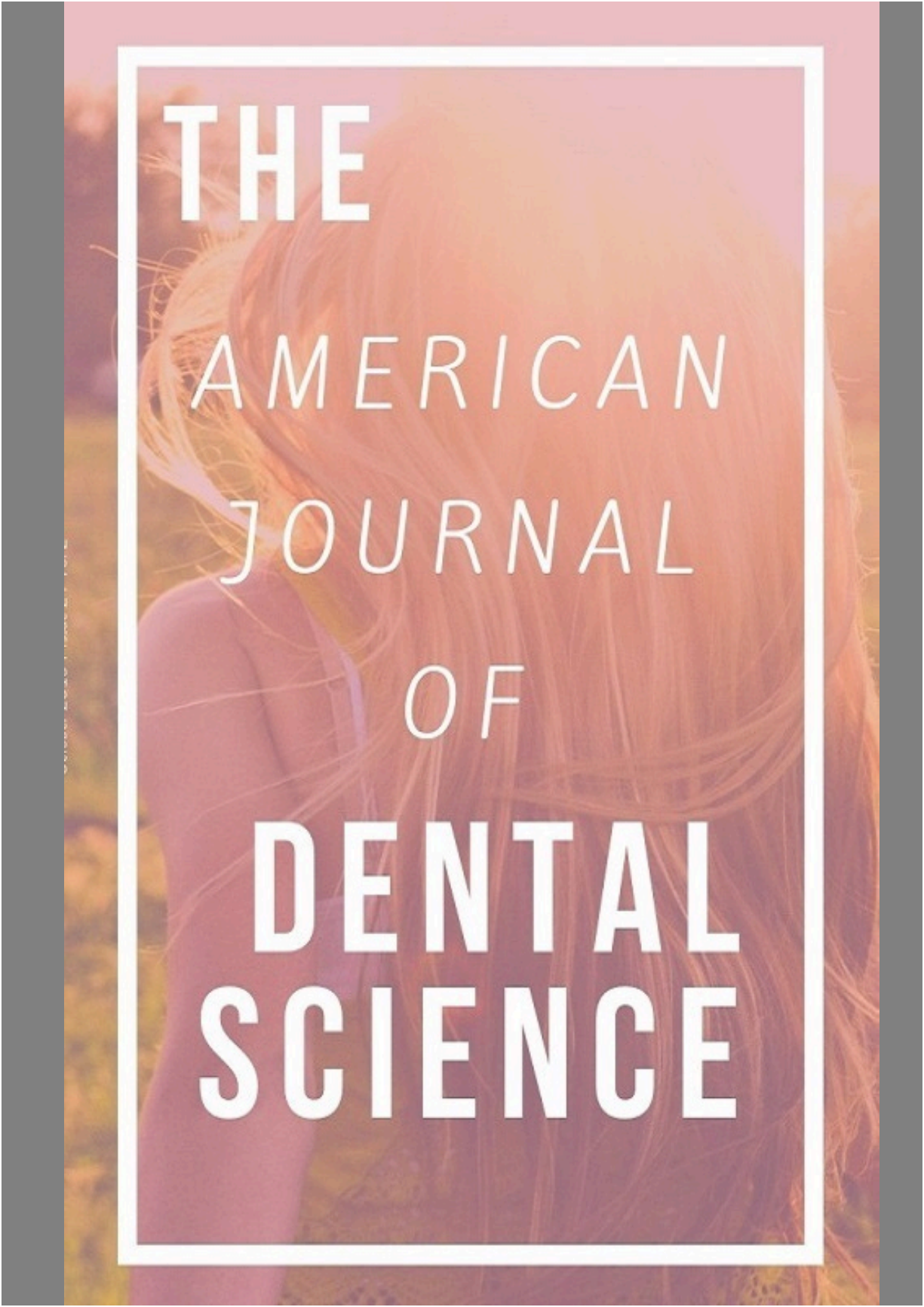
THE

AMERICAN

JOURNAL

OF

DENTAL
SCIENCE

The cover features a photograph of a person with long, wavy, light-colored hair, seen from behind, against a warm, golden-hued background. The entire image is framed by a white border, which is itself set within a larger grey frame. The text is overlaid on the central image.

THE

AMERICAN

JOURNAL

OF

**DENTAL
SCIENCE**

The Project Gutenberg eBook of The American Journal of Dental Science, Vol. XIX. No. 6. Oct. 1885

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: The American Journal of Dental Science, Vol. XIX. No. 6. Oct. 1885

Author: Various

Editor: Ferdinand J. S. Gorgas

Release date: August 26, 2019 [eBook #60187]

Most recently updated: October 17, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/60187

Credits: Produced by Jwala Kumar Sista and the Online Distributed Proofreading Team at <http://www.pgdp.net> (This file was produced from images generously made available by The Internet Archive)

*** START OF THE PROJECT GUTENBERG EBOOK THE
AMERICAN JOURNAL OF DENTAL SCIENCE, VOL. XIX. NO. 6.
OCT. 1885 ***

Transcriber's Notes

1. Typos were silently corrected.
2. The "cover-page" is developed and placed in public domain as well "Table of Contents" added by the Transcriber.

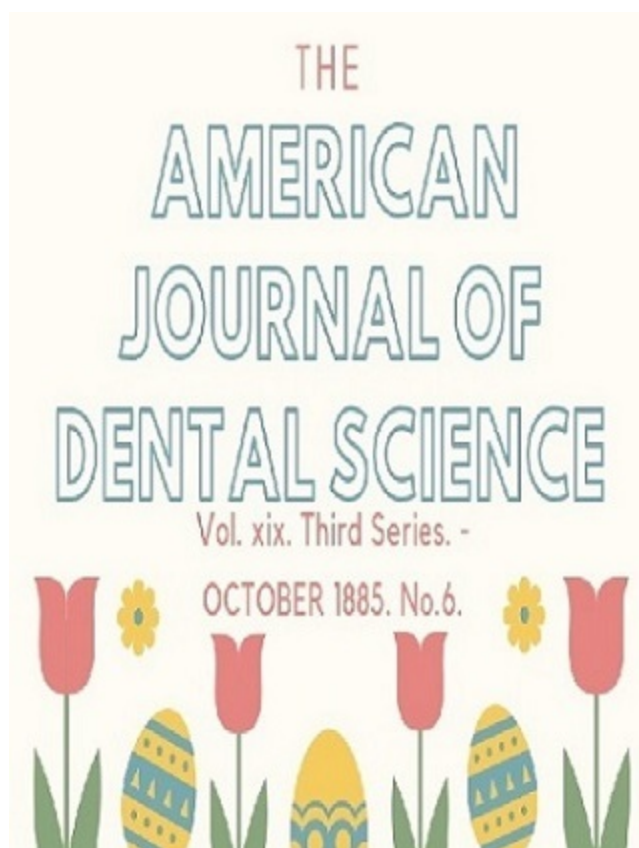


TABLE OF CONTENTS

[ARTICLE I.](#)
[ARTICLE II.](#)
[ARTICLE III.](#)
[ARTICLE IV.](#)
[ARTICLE V.](#)
[ARTICLE VI.](#)

[ARTICLE VII.](#)
[ARTICLE VIII.](#)
[Editorial, Etc.](#)
[Bibliographical.](#)
[Monthly Summary.](#)
[Footnotes](#)

**THE
AMERICAN JOURNAL
OF
DENTAL SCIENCE**

VOL. XIX. THIRD SERIES.—OCTOBER 1885. No. 6.

ARTICLE I.

NERVOUS ENERGY.

BY DR. E. PARSONS, SAVANNAH, GEORGIA.

[Read before the Georgia State Dental Society, May, 1885.]

GENTLEMEN—The subject I have chosen for your consideration at this, our Annual Meeting, is "Nervous Energy, how Actuated, and its Varied Phenomena." No one can question the importance of knowing all that can be known about it.

There is an invariable law by which means mind acts on matter, and it is my purpose, in this paper, to briefly elucidate what I have learned by reading, observation and experience on the subject. The great advantage of meeting in council is an increase in knowledge on all subjects in any way relating to our profession. We have many things yet to learn that will be, when known, of great benefit to both ourselves and the public.

Science demands a full and free investigation of all or any causative principle by which life is manifested, or death produced. So long as we draw our conclusions only from appearances, we shall often be deceived in a correct diagnosis; consequently often fail to cure diseases that come within the legitimate bounds of our specialty.

Proper remuneration for our services are absolutely necessary for the respectable maintenance of ourselves and those dependent on us; but our best men are laboring unweariedly in their endeavor to elevate our standard throughout the world, but particularly in our own country, and this Society can do much to help them in their onward march, developing all possible improvements in Scientific Dentistry.

Again, we all have a full consciousness of three things—we love, we think, we act. But few have a scientific knowledge of the means employed by which mind acts on matter. There are such varied forms and

circumstances controlling its development, that we need not marvel at anything that comes within the sphere of our observation. As we are brought face to face with almost every possible condition of the nervous system, our opportunities for investigating the various manifestations of nervous energy, its source and supply, may we not equal any other specialty in solving the great problem of cause and effect manifesting nervous energy? As the brain is the seat of all sensation, I briefly present some of the best authenticated views of its organization. I think you all will agree with me that it is wonderfully constructed by Infinite Wisdom for the development of the finite mind. In the elucidation of my subject, let us not forget the fact that the blood has much to do with the various conditions of the nervous system. It holds, or should contain in solution, all the elements necessary for the growth and sustenance of every organ in the body; it is both a receiver and a giver; it is fed from what we eat and drink, without which it cannot perform the office intended.

THE BRAIN.

The brain is divided by the septum into two lobes, right and left sides; also, into the cerebrum and the cerebellum, front and back sides. Phrenologists divide the lobes into about forty convolutions, assign to each a distinct office, and by careful observation of each as to their development, profess to know individual character, and point out what kind of occupation one, by nature, is best fitted for.

In 1840, Dr. Sherwood, of New York, by ingenious experiments, demonstrated the fact that the brain has four large poles, two in the cerebrum and two in the cerebellum, and from these proceed not only the convolutions, but every nerve in the body. By these, and experiments in animal magnetism, he maintained that animal magnetism is the motive power of the human system, and without it there can be no connection between mind and matter.

In the Fall of 1844, I invited several of our most eminent physicians to meet me at my office to witness some experiments in magnetism. My subject for demonstration was a young man of unimpeachable character, twenty-three years of age. My visitors were very skeptical on the subject of

magnetism. There had been some public exhibitions, but the result did not satisfy them.

I said, as the science of phrenology is ridiculed by some, I wished first to exhibit each convolution of the brain, in a state of exaltation, while he is as wide awake as we are. To prevent any suspicion of collusion, I handed them a chart containing the names of the different convolutions of the brain, and requested them to write on paper any question or the name of the organ, for me to excite.

The first paper had on it mirth. I placed the point of my finger over the organ, and he immediately broke out into an uncontrollable fit of laughter. I passed off the influence and he instantly became calm. They asked him what he laughed at. He said he did not know; he could not help it.

The second paper had on it veneration. I excited the organ, and he immediately bowed his head and assumed the attitude of the most zealous pietist, and appeared to be in earnest prayer.

The next paper had on it music. I excited the organ, and he immediately commenced singing with as much earnestness as if his life depended on it.

The next paper had on it combativeness. I excited the organ; he immediately doubled his fist and pitched into an imaginary enemy in the most vigorous manner possible.

Not to take up too much space in this paper, I will only add, we went through with about twenty of the organs with equally marked results, which both pleased and astonished my friends. They said they now thought there was much more truth in the science of phrenology than they had supposed possible. I then said you have seen the effect of so-called animal magnetism; I will now exhibit a different phase of it. I magnetized him in the usual way, and said he is now as oblivious to all external impressions as if his five senses had no existence. Examine him and satisfy yourselves.

After a thorough examination, Dr. Richardson said he believed he could cut off his leg and he would not feel it. I demagnetized him, which restored him to full consciousness. They questioned him about it. He said he did not remember anything done in that state. They then said if it was

practicable it would be a good thing in surgical operations. They thanked me for the pleasure of witnessing the experiments, and retired.

To understand the different nervous conditions of patients is of vast importance to both dentists and physicians. This cannot be attained without close observation and experience. If the nervous temperament of a patient is known, we shall have a key to guide us in our treatment in every individual case. Temperament is usually divided into six distinct classes:

1st. Nervous bilious; 2d. Nervous sanguine; 3d. Nervous lymphatic; 4th. Bilious nervous; 5th. Sanguine nervous; 6th. Lymphatic nervous.

You have doubtless observed a difference in the quality of human teeth. We usually find the best in the nervous bilious temperament, and the poorest in the lymphatic nervous. Viciated tastes and habits do not change the shape of the teeth when once formed, but their quality. Science demands discrimination under varied circumstances; it is not possible to treat all alike, and be equally successful. I shall refer to this again below.

I now present you with a few incidents in practice which may serve as a basis for a better elucidation of my subject:

1st. In the Fall of 1835, I was called to see a gentleman at eight P. M., represented to be suffering greatly, and unable to come to my office. I was introduced to a large man walking the floor in great agony. Seating him in a chair, I found the left side of the face swollen; a purple colored spot over the antrum; the first molar on the left upper jaw filled with gold; tooth firm, but evidently devitalized. I diagnosed the trouble to be abscess in the antrum. I extracted the tooth; he sprung out of the chair and dropped on the floor, face downwards, and quivered like an ox struck on the head with an ax. I used cold water freely to his head, and soon brought him to; placed him back in the chair; made a free passage through the front labial socket into the antrum, and the pus flowed freely. He then laid down on the bed much relieved; gave him a half grain of opium; waited about twenty minutes and injected the antrum with warm green tea; directed his head to be kept cool with cloths wet with cold water; left him with a promise to call again in a few hours; called again about three P. M.; was told he had been sleeping several hours; I injected the antrum with a weak solution of nitrate of silver; saw him the next morning and again injected the antrum with a much stronger solution of nitrate of silver; the swollen cheek appeared

almost natural; said if he needed my service any more to come to my office. About a week later he called on me; brought me a sack; said a few hours before that he blew it out through the left nostril; it was about one and a fourth-inch long, and about the size of an ordinary goose-quill; it was soft with a leathery like appearance; said he was all right; paid his bill; have not seen him since.

2d. In September, 1836, I was called by a physician to see his wife. He said she was suffering terribly from facial neuralgia, and thought it was caused by a tooth. She was in her eighth month of pregnancy. He had applied the usual remedies, which gave no relief. I examined her teeth; found both the third-molars on upper jaw decayed, and on slightly tapping them with the handle of an instrument, the pain was greatly increased. He said he was afraid of the consequences, in her state, of having the teeth extracted. I told him nothing else would give relief. She said take them out, it cannot be worse than I am now suffering. I parted the gums from the teeth with a lance; I had barely completed this part of the operation when she fainted; her mouth was open; I took my forceps and extracted both teeth; brought her head forward to prevent the blood running down her throat; we soon brought her back to consciousness, and the first thing she said was: "I cannot have them out." Her husband said: "Darling, they are both out." She said, "Are they? I did not feel it; I am so glad; the pain is all gone." The Doctor said to me—"You are a bold man: I would have stopped you if I could, but you was too quick for me." I afterwards learned that no serious result followed the operation.

3d. In August, 1856, a young lady, aged about twenty, came to my office at eight o'clock A. M. Temperament nervous, lymphatic. Said she had not slept a wink all night. Her face was pale, hands cold, pulse feeble. Said she had a mortal dread of having a tooth extracted. I put my mouth-mirror into her mouth for examination, and saw the tooth caused the trouble. In an instant she fainted. I took my forceps and extracted the tooth, used restoratives and soon brought her to. The first thing she said was—"I cannot have it out." I showed her the tooth. She said: "Oh! I am so glad I did not feel it;" and left the office laughing about it.

4th. In the Fall of 1858 a lady called to have the two upper front incisors filled. She appeared to be middle-aged, and apparently in good health. On examination, the teeth were very close together—not badly

decayed, but must be separated for sufficient room to enable me to do the work properly. Our only means, then, was either wedging or filing them. I filed about one quarter of what was necessary, and she fainted. I then filed as rapidly as possible while she was unconscious, and completed this part of the operation, and used restoratives, and soon brought her to. I gave her a glass of wine, and completed the operation without further trouble.

5th. In the Fall of 1874, a lady called to have a tooth extracted. She appeared to be in good health; said she was almost distracted with toothache; was afraid to take chloroform; was afraid as of death without it. I said the pain would be only momentary, and would not kill her. I extracted the tooth, and she fainted. My usual remedy in such cases was hartshorn and cold water. Through mistake, I took up a vial of the Essence of Gaultheria, poured a little on a handkerchief, held it to her nose, and was surprised to see how quickly she recovered consciousness. This prompted me to experiment with it. I concluded that if it was a good restorative, it might be a useful preventative.

I soon had a chance to test it. A lady called to have an ulcerated tooth extracted. She was in delicate health: face swollen, hands cold. She said she would like to take chloroform, but her physician said she must not take it; she knew she would faint without it. I told her I thought that could be prevented. I took a doily, folded it small, and poured about a teaspoonful of the essence on it. I told her to inhale through her nose, and exhale through her mouth. She continued this until her brain was pretty well stimulated, and the tooth extracted. She showed no signs of syncope, and could hold a glass of water as still as I could. I have not had any one to faint away in my office since.

In the Spring of 1874, a lady, aged about sixty, came to consult me. She said that her teeth were so bad she could not eat any ordinary food; had disease of the lungs; was forbidden to take chloroform. After an examination, I told her she had eight teeth in the upper jaw that could not possibly be made useful, and she had better have them extracted. She said she had never had one extracted without fainting dead away. She could not think of having more than one out at a time. Her temperament, nervous sanguine, emaciated hands cold, pulse very feeble. I told her if she would follow my directions I would take them all out and she would not mind it more than one, and guaranteed she could not faint if she tried. I explained

the effect of the wintergreen, and said it would do no more harm than a glass of good wine. I administered the article, as before described, until her face flushed, tears ran down her cheeks, then extracted the eighth teeth without her closing her mouth. She asked if they were all out, and I said yes. She said, "Is it possible?" I gave her a glass of water. It did not show the least tremor of the nerves. She left, giving me many thanks, saying that she felt much better than when she came into the office.

I could relate many more similar cases, but do not deem it necessary. My object is to show what may sometimes be done to advantage in cases of syncope, and also the means of preventing it while performing a painful operation.

As before said, the brain is the seat of all sensation; and our patients, no matter how nervous they are, if the brain is properly stimulated, cannot faint—caused by the extraction of teeth. When my patients are known to be pregnant, I always use the stimulant above described before performing any painful operation; it always prevents any severe shock of the nervous system when in this condition. If any one wishes to know how to prepare the Essence, it is as follows: To one pint of alcohol add one ounce of the Oil of Gaultheria, commonly called Wintergreen. Shake it well and it is fit for use.

In passing from a conscious to an unconscious state, all the Clairvoyants I have questioned on the subject say it is affected by a change of polarity of the sensory organs, and the principle is the same whether caused by animal magnetism, syncope or anæsthetic agents, and if only the voluntary organs are affected thereby, there is no danger to life, but if polarity in the involuntary is reversed, the heart ceases to beat and death is instantly the result. I was the first in this city to administer ether for the purpose of extracting teeth without pain. In a few cases it developed paroxysms of hysteria; otherwise no harm was done. I have administered ether, chloroform and gas to over two thousand persons. With chloroform, I had three cases that barely escaped death in my chair; with gas, some after deleterious effects followed in two cases.

Admitting man's physical organization to be a magnetic machine, the deaths that have occurred are easily explained, when caused by these powerful drugs. The voluntary organs are under the control of the will, and

during our waking hours there is a constant draft on our magnetic supply; it is best recuperated by sleep, when the will is at rest.

The involuntary organs do not sleep until death ends our earth life. We can readily understand that if by any cause polarity in the two large poles in the cerebrum are reversed, the gateway by which we gain a knowledge of things about us is closed so perfectly that physical sensation is impossible. On the other hand, if the equilibrium between the two large poles in the cerebellum are not well balanced, just in this proportion some kind of ailment is the result. Let us not forget that the will has no control over these poles, and all medicine that does not beneficially act on them is non-curative. Now, just in proportion as anæsthetic agents disturb their equilibrium, they are dangerous, it makes no difference whether polarity is reversed or destroyed; in either case the principle of life can no longer act on the nerves by means of its intermediate; the heart ceases to beat, and restoration is impossible. Chloroform is more easily administered than ether or gas, and most convenient when the patient cannot come to the office; but we should remember that many deaths have occurred when given for the purpose of extracting teeth, and that, too, when least expected. The public mind is more horrified at one death in the dentist's office than twenty caused by a railroad smash-up. We have now the means that will stimulate the nerves, greatly mitigate the pain and not endanger either life or health.

I have long desired a perfectly safe anæsthetic that can be administered no matter what the condition of the patient. I am now creditably informed that Dr. Mayo, of Boston, some eighteen months ago, by various experiments, produced a compound article that satisfied him was harmless. He would not put it on the market until it had been thoroughly tested by both dentists and surgeons. All who tested its effect and efficiency testified to its great superiority over all other known anæsthetics for dental and minor surgical operations. It is now only a few months since he made arrangements for its manufacture and appliances, and put it on the market. He has named it Mayo's Vegetable Vapor Anæsthetic. I have been using it for extracting teeth very successfully. The nitrous oxide causes the patient, when fully under its influence, to have very like the appearance of a corpse. The action of this new anæsthetic does not act on the vital organs, and the patient appears like one in a natural sleep, and, in my opinion, is perfectly safe and without danger to life or health.

Our patients come to us for either a preventative or curative treatment. As before said, we are brought face to face with almost every conceivable condition of the nervous system, and the more true knowledge we have of it the better are we able to satisfactorily manage them. Some come in a very excitable, and some in a very depressed state. We need means to quiet the former and stimulate the latter.

Again. To be fully entitled to the name of Scientific, we must know something of the laws of life, in order that we may obey them and fight life's battles manfully—doing justice to others and with credit to ourselves. Life, in itself, is not creatable, but given to us with power to properly use or abuse, the end being the creation of the finite mind. I have explained above the means by which it acts on matter, but as a further illustration, let me say, you drop an article on the floor, gravitation holds it there; you desire to pick it up, how can you do it; if you have sufficient will-power it will act on the magnetic element, this on the nerves, these on the muscles. You stoop down and pick it up, and probably not one in ten thousand have a single thought about the necessary means by which you are enabled to do so, so little do we reflect about causative principles involved in what we do.

So far as our voluntary organs are concerned they may be compared to a locomotive engine. They are both useless if the motive power is wanting. To make the engine useful, steam must be generated by means of fire and water; and to make our voluntary organs useful, animal magnetism must be generated by means of life and the atmospheres. The engineer controls the steam power, and human will controls the magnetic power, and when properly applied, if the machine is in good order, locomotion is the result in both cases. I will only add, the steam acts on the piston heads and causes the crank to move and the wheels to rotate. Magnetism acts on the nerves, then on the muscles, and man moves in any direction he chooses. A dentist with a strong will, if he uses the proper means, can more easily and favorably impress his nervous patient than one with a weak will, and the reason is, he imparts more of his animal magnetism, which has a stimulating effect on the nerves of his patient. There is a magnetic sphere emanating from both man and beast, particularly when in motion. Were it not so, no dog could follow their tracks successfully. All pain is the result of an obstruction of a normal flow of the magnetic current, whether caused by disease or otherwise.

Arsenic, applied to the nerve of a tooth, destroys its polarity, and applied to any other nerve it is no longer capable of being actuated by the magnetic current, without which there can be no sensation, and death is the result. Physical endurance depends largely on the mind and the state of the nervous system. The difference in individuals to bear pain is marvelous; some one can have a tooth extracted and seem to care but little about it, while others, without the use of a preventative, appear to suffer intensely, and in some cases the operation causes syncope.

In conclusion allow me to say, a vast field lies before us, and if cultivated properly this Society will in due time reap a rich harvest, the benefits of which cannot now be estimated.

The grand distinction between mind and matter may be seen thus: If we give any physical object to another, we part with it, but if we give a new idea on any subject, we do not part with it, but in so doing its boundaries are enlarged in our minds.

The space occupied in briefly presenting my views on the subject I have chosen is greater than I at first intended, but the fundamental principles which underlie everything with which we have to do, and the importance of fully understanding them, is my only apology for occupying so much of your time.—*Dental Luminary*.

ARTICLE II.

PULPLESS TEETH.

**BY DR. WILSON, OF BURLINGTON, MEMBER OF THE FACULTY
OF THE DENTAL DEPARTMENT OF THE STATE UNIVERSITY.**

[Extracts of a paper read before the Iowa Dental Association at its late annual meeting.]

A pulpless tooth is not necessarily a dead tooth, but a dead tooth is, of course, a pulpless tooth. The adjectives "pulpless" and "dead" are not, therefore, synonymous, although frequently so used, especially by medical writers. Let us note the marked distinction between the two. A pulpless tooth may be a part of the living organism—a dead tooth has its nutritive supply entirely cut off, and it is in every sense a foreign body—it is dead and inert. The former may be restored to health and usefulness—the latter should always be condemned as a nuisance that cannot be abated without the use of the forceps. * * * *

Having thus briefly called attention to the fact that the dentine and cementine derive their vitality from independent sources—that the life of the one is not dependent upon the life of the other—that a pulpless tooth is not necessarily a dead tooth—we are prepared to consider, understandingly, the subject of this paper. It may, however, seem like presumption on the part of the writer, in thus offering the foregoing to an intelligent body of dentists, when every student of dentistry at the close of his junior year should fully understand the facts above stated. But I am led to a consideration of this subject from articles entitled, "Dead Teeth in the Jaws," that have appeared, from time to time, during the last two years, in the *New York Medical Record*, and as those articles come from high sources in the medical profession, they deserve more than passing notice. The able editor of that journal, and Dr. Samuel Sexton, a distinguished oculist and aurist of New York City, being the principal writers referred to.

The *Medical Record* of October 4, 1884, contains a report from the aural service of Dr. Sexton, entitled, "Pain in the Ears due to Irritation in the Jaws." He describes a number of cases of otalgia in which he found the lesion to be in diseased teeth.

He goes on to say that "since dentistry had become such a popular business, and diseased teeth had been so carefully retained in the jaws, nervous diseases about the head were becoming alarmingly common."

The same number of the above journal contained an editorial on "Dead Teeth in the Jaws," which read as follows: "Perhaps the time is near at hand when medical men should be themselves better informed concerning diseases of the jaws and mouth, rather than refer the ailments of this region to individuals whose limited knowledge of medicine does not prevent them from 'treating' dead teeth long after their presence in the jaws has given rise to alveolar abscesses and neuralgias more or less painful. It would not be strange if in the course of events, the day would soon come when all teeth without pulps, and hence in process of more or less rapid decay, as well as those which the deposit of tartar, or other cause, had become entirely divested of periosteal nourishment, would be promptly condemned as unfit to remain in the jaws, regarded in fact as foreign bodies liable to give rise, not only to cerebral irritation and disease in the organs of special sense, through the propagation of local disturbances in the mouth to the regions mentioned, but to endanger likewise the general health through purulent matter discharged into the mouth from alveolar abscesses, to be continuously swallowed for a long time, or, indeed, in some instances, to be absorbed and thus produce septicæmic poisoning. It is certainly gratifying to note the establishment of instruction in oral surgery in some of the medical schools, and it is to be hoped that this subject will receive the attention its importance demands."

Dr. Sexton cites the readers of the *Record* to eight cases of otalgia resulting from diseased teeth. I have no doubt but a majority of the dentists before me to-day have met with almost that number of cases in practice every week; nor do you find it a difficult thing to render prompt relief, and that, too, in a large majority of cases, without the use of the forceps. And I believe that I am warranted in saying that in at least three-fourths of the cases met with in our practice, we find the reflex pain in the ears due to exposed living pulps, and not to "dead teeth in the jaws."

That diseased teeth do cause reflex trouble, not only in the head, but frequently in more remote parts of the body, is a fact well-known to every competent dentist. I am glad that Dr. Sexton has at last discovered the fact, that diseased teeth do frequently cause reflex pain in the ears, and in other neighboring parts, and that alveolar abscesses very often cause catarrhal affections of the maxillary sinus and of the nasal passages, and that diseased teeth will endanger the general health. It is to be regretted, however, that the doctor has found it necessary to charge this unfortunate state of affairs to the ignorance of dental practitioners, who are in no way responsible for but few of the many cases met with in practice, for there can be no doubt but a very large majority of the teeth causing the troubles above referred to have never received any treatment whatever at the hands of dentists, and because Dr. Sexton has discovered that in certain cases pulpless teeth (or dead teeth as he calls them), has caused the ailments above referred to by Dr. Sexton, there can be no doubt. Every dentist of any considerable experience can enumerate such experiences by the score, and the medical profession has only been too slow to recognize the facts discovered by Dr. Sexton.

The only difficulty with these medical gentlemen is, that they have drawn very erroneous conclusions from the important discoveries they have made. Their limited knowledge of the minute structure of the dental tissue, and the source from which each derives its life, is manifested by the erroneous statements upon which they have based their arguments, and then after arguing from false premises, Dr. Sexton says: "In regard to the treatment of pulpless teeth, the practice in vogue seems the reverse of procedures founded on well-established surgical principles." And in an editorial of the same issue we are informed that the treatment of diseased teeth is carried, to what "the medical minds regard as a dangerous extreme."

That some members of our profession have been over zealous in their efforts to save all diseased pulps alive, there can be no doubt. We will occasionally meet with an enthusiast in our profession who will say, "I have no use for forceps, I never extract teeth." I have heard that statement made on the floor of the Iowa State Dental Association.

That incurable diseased teeth should not be tolerated in the jaws does not admit of discussion. Good common sense ought to settle that question. And again, there are extremists who never devitalize diseased pulps, no

matter how badly exposed, but "doctor them up," and stupify them, and then bury them in a living grave. Much evil has grown out of this practice.

Some one has said that to cap a badly exposed pulp is to create a slumbering volcano, and he might well have added that such volcanoes have but a limited time to slumber. Gentlemen, there are in our own country ten thousand volcanoes belching forth—not pure molten lava—but impure gases and putrescent matter of the most sickening character. The craters to these volcanoes are not found on the mountain top, but they are found in human mouths—in the antrum of Highmore, in the nasal passages, and externally on the face, neck, or even on the chest.

When the pulp of a tooth is dead and confined within its bony walls an outlet is sought, and must be affected for the escape of impure gases arising from the decomposing pulp and for the putrescent matter associated with it. When thus confined its only way of escape is through the dental foramen, and into tissues adjacent thereto. The pressure thus brought to bear upon the bony walls surrounding the apex of the root will in time perforate it at its weakest point, and the poisonous matter is forced through the opening thus formed and into the soft tissues, which soon yield to the pressure, and the imprisoned mass of corruption is liberated. The pain and swelling now subsides, but a dangerous nuisance has been created. The channel formed from the apex of the root to an external opening will not close while it is used for the passage of foul matter and gases that will flow unceasingly from the pulp canal.

The remedy of course is to remove the cause, and assist nature in affecting a cure, and to do this the pulp chamber must be opened, its contents removed, the canals cleansed and disinfected, the abscess healed, and the roots filled to the exclusion of all fluids and purulent matter. But how often this is not done. How many thousands of suffering mortals are to-day dragging out miserable lives because of these drainage tubes emptying themselves into the oral cavity—into the maxillary sinus or into the meatus of the nose. Such an abiding nuisance in the mouth cannot long exist without ruining health. But how few of the unfortunate sufferers realize the cause of their nervous irritability, their loss of appetite, their feeling of lassitude, their lack of energy, and their general prostration. And here let me say, that but few, in comparison to the number of these unfortunate sufferers seek relief at the hands of the dental practitioner. The patient is neither sick

nor well, but debilitated and "good for nothing." The family physician is consulted, nervines and tonics are administered, but to no avail. The septic matter is vitiating the air that is breathed, and poisoning the food that is eaten. The saliva that is poured into the mouth from the various glands must mingle with this poisonous matter and carry it into the stomach.

Sanitary means are being employed in all our cities at the present time, in view of the cholera scourge that it is feared will sweep over our land the coming summer. Our physicians wisely talk and write about the baneful influences of impure water, about miasma arising from the decomposition of vegetable matter, and about unwholesome food, and it would be well if the public would heed their timely warnings. And as dental practitioners, I feel that we, also have an important duty to perform, in enlightening our patients, and the public so far as we are able to do so, in the direction I have above indicated.

The subject is of paramount importance, and as the opportunities come to us in every day practice, let us not fail to impress upon the minds of our patient (when we find it necessary to do so), the fact that a clean mouth is essential to health.

The agitation of this subject, by the medical profession, is a step forward. Hitherto medical men have not given the matter the attention its importance demanded.

And now that this new light has dawned upon Dr. Sexton, it is not strange that, in hastily drawing his conclusions, he should have mingled much of error with the truths he has discovered. Possibly some of the cases that have come under his notice may have been the result of bad practice on the part of incompetent dental practitioners, but to charge the dental profession with their short-comings would be a matter of great injustice. Dr. Sexton is too hasty in his conclusions. First, he discovered that certain pulpless teeth had caused certain ailments, hence he condemns all pulpless teeth. He has discovered that certain dentists have failed to treat such teeth successfully, hence he condemns the dental profession for attempting to save teeth, it would be equally fair to condemn the whole medical profession, because of the incompetency of some of its members. But before dismissing the subject of pulpless teeth, it may be well for us to examine the subject a little more carefully from the standpoint of the

medical writers above referred to. We cannot afford to make a mistake with regard to so important a matter. The higher a man stands in his profession, the more serious the mistakes he makes, and the more important it is that his practice be sound. An enthusiast or an extremist may injure a good cause. There are such men in our ranks.

A few years ago a prominent dentist said, "The tooth's pulp is its soul, and it is criminal to destroy it."

I heard another prominent dentist say, "If I find a part of the pulp dead, I amputate the dead tissues, and save the balance of the pulp alive."

A dentist has just moved away from Burlington, who has been in practice there for fifteen years, and during that time he has been using arsenic for obtunding sensitive dentine, and he has succeeded in accomplishing his purpose admirably. I have found in one month half a dozen filled teeth containing dead pulps, and, of course as many alveolar abscesses in active operation. The evils arising from such abominable methods of practice are simply appalling.

I have less frequently met with cases where those fistulous openings were on the neck or chest. In those cases the roots of the teeth are usually long, and when the abscess breaks through the lower border of the jaw, and the pus comes in contact with the soft tissues, it follows the course of the muscles and forms a sinous as it gravitates to some point on the neck or chest. I have known of a number of such cases being under medical treatment for years, where the affection was supposed to be of a strumpous nature, and the real cause was not suspected, and in every case a rapid recovery has followed the extraction of the offending tooth.

Gentlemen, I have no doubt but the most of you are disappointed in the nature of this paper. I have scarcely alluded to the treatment and filling of pulpless teeth. That had not been my purpose. But I have wished to call attention to the fact that a large majority of the ailments above referred to have been due to diseased teeth that have never received any attention whatever at the hands of competent dentists.

That pulpless teeth and roots may be treated, filled, and preserved in health in a majority of cases, is a settled question. Every well-informed

dentist knows that to be a fact, the distinguished Dr. Sexton and the able editor of the *Medical Record* to the contrary notwithstanding.—*Iowa State Med. Reporter.*

ARTICLE III.

DEAD TEETH IN THE JAWS.

TRUMAN W. BROPHY, M. D., D. D. S.

In reply to Dr. Sexton on this subject, Dr. Brophy makes these pertinent remarks in the journal of the American Medical Association:

Dr. Sexton says: "The retention in the jaws of teeth which are diseased, have become irredeemably sensitive to thermal influences, or deprived of adequate periosteal nourishment through calcareous formations about the roots, very frequently gives rise to nervous diseases about the head. I am convinced that these reflected nerve influences manifest themselves much oftener since dentistry has come more extensively into practice during the present generation, and greater efforts are made to retain defective teeth in the jaw."

That diseases of the teeth are often the center from which pain is reflected to the eyes, ears and other parts, all experienced clinical observers must admit. But that these pathological conditions of the teeth, from which reflected pain has its origin, can be and are successfully treated and cured with rare exceptions, as effectually as any other diseases, is a fact too well established to be set aside.

It is not possible to describe in this letter the method by which the various diseases of the teeth are treated, but suffice it to say that "teeth which are diseased from death of the pulp or from caries" *do not* "become irredeemably sensitive to thermal influences." In proof of this statement, many thoroughly educated medical men, practicing the specialty of dental surgery, will testify.

"Teeth deprived of adequate periosteal nourishment, through calcareous formations about the roots, very frequently give rise to nervous diseases about the head." To this statement I assent, but dissent as to the

remedy not mentioned but implied, *i. e.*, the removal of the teeth. If the calcareous deposits mentioned have destroyed so much of the pericementum and the alveolar processes as to render the teeth very loose; if, indeed, the teeth have lost their bony support and are retained by means of a remnant of pericementum only, they cannot, of course, be restored to permanent health and usefulness, and their removal is, therefore, indicated. Teeth in this condition "frequently give rise to nervous diseases about the head."

On the contrary, if the calcareous deposits have not destroyed the pericementum and alveolar processes to a very great extent, the condition is amendable to intelligent treatment and cure. In answer to the assertion that "Reflected nerve influences manifest themselves much oftener since dentistry has come more extensively into practice during the present generation," I would say, that with equal propriety it might be said that reflected nerve influences manifest themselves more frequently since gynæcology has come more extensively into practice. To attribute the obvious increase of nervous diseases during the present generation to diseases of the teeth is a statement not only "sweeping," but "overdrawn." *Much harm* is no doubt done by some of the modern appliances "for retention in the mouth of substitutes for absent teeth," and the unhealthy state of the gums and contiguous parts, established and maintained by the presence of these substitutes, unquestionably give rise in many cases to reflected pain.

When Dr. Sexton attempts to establish a *law* governing the management of diseased teeth, it must be based on more substantial grounds than those which he presents. The case related of his patient, the "medical man, who practices dentistry," and who was convinced that an inflammation of one of his ears began from the time the upper second molar of that side was treated for a diseased pulp, is simply an assumption, on the part of the patient, that the ear trouble had its origin from the diseased tooth, and the patient's diagnosis of his own case seems to have been accepted by Dr. S. as conclusive. The ear disease in this case may have emanated from the diseased tooth, but no evidence is produced to that effect. In regard to the query as to "whether it is safe practice to retain dead teeth in the jaws," I would say that thousands of people in our own country have had pulpless (not dead) teeth in their jaws many years, which are exempt from

pericemental disease, and which serve all the purposes for which teeth were provided. To ask whether it is safe practice to retain these, so-called, dead teeth in the jaws when they have been comfortable and useful from ten to forty years and promise to remain so through life, seems like a proposition too injudicious to need comment. While the death of the pulp results in "cutting off the source of nutrition from the dentine," it does not follow "that in a large number of instances irritation can not be easily controlled."

Neither does the tooth become a foreign substance. The dentine and the enamel are, of course, no longer nourished after the death of the pulp, but their resisting structure renders them capable of maintaining their integrity many years after the pulp has been removed; and pericementum will nourish the cementum and thereby retain the tooth in its alveolus in a comfortable condition. In order, however, to thus retain the tooth and prevent inflammation from supervening, the devitalized pulp must be removed, the pulp canals thoroughly disinfected and filled with a plastic material which hardens when in position. Dr. S. most clearly exhibits his imperfect knowledge of the dental operations in vogue when he says: "Inflammation of exposed dentine cannot surely be entirely arrested in any case by filling the pulp cavity with any known extraneous material, and especially is handicraft wanting to even imperfectly protect the minute and often tortuous canals leading down to the apical foramina of the majority of the teeth." To arrest "inflammation of exposed dentine by filling the pulp cavity," in the opinion of Dr. S. would seem to be most desirable. How a tissue without nourishment and consequently without vitality can take or maintain inflammation is beyond comprehension. The impervious filling which I have mentioned will close the apical foramina, together with the canal, which "in the majority of cases" is *not* tortuous to a degree of rendering the perfect filling of the root difficult or uncertain, and the assertion that the dental surgeon "is able only to offer a hopeful but uncertain prognosis in these cases" is contrary to well established fact. There are no diseases to which mankind is heir more scientifically and effectually cured than the diseases of the teeth in question.

Again: "The dead tissues of the dentine will sooner or later, most likely, be transmitted through the tissues of the cementum to the periosteum." Communication between the lacunæ canaliculi of the cementum with the tubuli of the dentine is not free; indeed, it seldom exists,

hence it cannot be "that through the periosteum alone the dentine may long derive some nourishment."

About 22,000,000 teeth are annually extracted in the United States, and I regret to say this enormous loss of teeth is to no small extent due to the indifference manifested by physicians in the anatomy, physiology and pathology of these organs. It is a fact, no one will attempt to gainsay, that hygienic measures directed toward the preservation of the deciduous set, if understood, are seldom recommended by the general practitioner to the families under his charge. The premature loss of these teeth paves the way for early lesions of the permanent set. The pain resulting from advanced caries of the deciduous teeth, owing to the difficulties encountered in controlling the patient, is not easily treated; moreover, the injurious impressions thus made on the system of the child abide through life. There is no doubt hundreds of thousands of teeth are unnecessarily extracted each year, and then drugs are given with a view of curing the patient of the disorders of digestion and other abnormal conditions which follow, and which in turn arise from imperfect mastication of food, verily for the want of teeth.

We need to know "what's the matter" in the treatment of these "nervous diseases about the head," as in all others, and apply a remedy which will bring the abnormal tissues back to health. Too often, indeed, has it happened that patients, by advice of their medical attendants, have submitted to the loss of many, and, in some instances, to all their teeth, in the vain endeavor to be relieved from trigeminal neuralgia. You may ask, Why this useless loss of teeth, and all the resulting evils? Because the advice given was not wise; the etiology of the affection was not understood.

There are certain pathological conditions of the teeth which have not been mentioned in this discussion, and which give rise to reflected pain of the eyes, ears, and other parts.

Among these may be mentioned exostosis of the roots of teeth and nodules of calcific matter within the pulp canals in contact with a living pulp. The former of these conditions has been regarded incurable, the removal of the tooth with the united bony tumor being indicated. In favorable cases, however, this tumor may be excised and removed without removing the tooth. The pulp nodules of calcified deposits within the pulp

chamber may be, in a large majority of cases, successfully removed without sacrificing the tooth.

No one approves more than I the removal of the causes of disease. It is no more necessary to extract a tooth at the root of which an alveolar abscess has formed than it would be to amputate a limb for the cure of an abscess of the medullary substance of its bone. Disease of the eye sometimes requires that it be enucleated, but the honest, skilled ophthalmologist *would not* remove the eye when he *knew* he could restore it to usefulness. The spirit of the teachings of Dr. Sexton's articles is far from being progressive. Nor is this all; many assertions are not based on fact, but on erroneous impressions. Our duty to our profession and the laity is not to destroy but to save; and while ignorance is ever working its mischief in all vocations in life, it is not just to accept the results of such work as a basis on which to found a law.

ARTICLE IV.

DIAGNOSIS AND TREATMENT OF DENTRITIC CYSTIC TUMORS OF THE JAWS.

BY JOHN S. SMITH, D. D. S., LANCASTER, PA.

Diagnosis.—Cystic tumors may be confounded with other affections which occasion swellings about the jaws, as enchondromata, sarcomata, and myxomata, abscesses, and the collections of fluids in the antrum. Dental alveolar abscess may be distinguished by its acute course, and when in a chronic, condition by the discharge of its contents through the fistula, either upon the gum, or within the oral cavity. The tumor formed by an abscess is never so sharply definite as is the case with cysts; with dropsy of the antral cavity the distention of the facial wall of the jaw is more uniform than it is with cysts.

In some cases of cystic tumors, they present so formidable an appearance at first sight, that they may be taken for solid tumors; especially is this so when their walls are compact and well organized, nearly if not altogether obliterating the sense of fluctuation when pressure is made upon them.

Cases have come under the observation of the writer where it required the most delicate touch to detect any fluctuation when pressure was made upon the apex of the tumor.

In some cases the diagnosis cannot be determined accurately until after one or more teeth are removed that are involved with the tumor. After such operation, a probe carried through the alveolus will usually reveal the true condition of the lesion. One or more dead teeth are found involved—one, however, being the rule in most cases which have come to the notice of the writer, while two, and sometimes three, are implicated with the the tumor. The dead tooth may be easily distinguished from the living ones by its opaque appearance. Such tooth may be carious, and it may not.

Primarily the dentritic cyst originates from what pathologists call a "cold abscess," that is, an abscess which has never opened; subsequently, having developed into a tumor. The interior of the cyst has a fibrous lining, and being compact in structure, is the seat of an inflammatory process. The cyst contains a pyriform fluid; it may attain such magnitude as to invest several teeth and extend beyond the alveolar process. The tumor is usually oval in shape, with its apex on a line with the diseased tooth directly involved. The size of the tumor may be as large as a hulled walnut or as small as hazel-nut; crepitates under pressure, and feels like parchment. In cases of long standing, considerable resorption of the alveolar process takes place, and the teeth immediately connected will be loose; especially will this be the case if the alveolar borders are broken; these teeth should be removed. These tumors are found painless, as a rule. I have met with cases, however, where an acute inflammatory condition was present, with all the symptoms of acute periodontitis manifested. So that it could have been readily mistaken for the pointing of an alveolar abscess.

Pathology.—Cysts of the jaw may be either simple or compound; whether they be cysts of retention, exudation cysts, or extravasation cysts belonging to the jaws, is a matter not as yet fully established. The exudation cyst is a secretory cyst; in a generic relation, however, it is just the opposite of the retention cyst. Serous sacs form the foundations of the exudation cysts. "The mode of development of cysts of the jaws," says Wedl, "has not yet been determined; it therefore becomes necessary, in order to throw more light on the subject, to pursue further anatomical investigations in that direction."

Rindfleisch says: "The accumulation of the fluid is not produced by the continuance of the normal secretion, but by an exudation surpassing the normal measure of the serum of the blood with salts, albumen, fibrinogenous substance, and extractives, in the most varying proportions. The exudation cysts have little to do with pathological new formation. Of extravasation cysts," he says, "a parenchymatous bleeding can very well be the point of departure for the formation of a cyst. The hemorrhagic depot can present itself primarily as a cyst, namely, when the blood is poured out between two surfaces in themselves smooth; for example, bone and periosteum, cartilage and perichondrium, and thereafter remains fluid. As a cyst may also be formed when upon the one hand the limitary parenchyma

furnishes a connective tissue membrane, upon the other hand, the blood itself is resorbed through a series of metamorphoses up to a small remainder, and is replaced by a clear fluid."

The above-mentioned condition is liable to manifest itself within the body of the jaw, the bone and periosteum, after severe mechanical injuries to the bone, and the rupture of blood-vessels within the parenchyma. There can be little doubt that many of the so-called dentritic cysts of the jaws have their origin primarily from causes brought about by falls, strokes and mechanical violence, causing rupture of blood-vessels. It is quite true, history of cases fully confirms such facts.

Clinical observations leads us to believe, however, that only in cases where the abscess does not open, we find the pathological new formation taking place within the jaws. Pulpitis, and as has been observed, followed by pericementitis and periodontitis, is a prolific cause of the development of the dentritic cystic tumor.

Treatment.—The removal of all dead teeth involved. Other teeth whose pulps are living may be loose, and to a casual observer appear to be complicated, but a careful examination will reveal the fact that they should not be disturbed but retained in their places; only one tooth may be the offender, being a dead one which has caused the trouble. After the removal of the cause, let it be either one or more dead teeth or fangs of teeth, cyst walls may be punctured with a sharp instrument, and the contents of the sac released, this being done by carrying the instrument through the alveoli, and not through the bony parietes of the jaw. After the contents of the sac is let out, and the sharp spicula of bone trimmed, with engine burs, tincture of iodine full strength may be forced into the cyst sac, by saturating tufts of cotton-wool and allowing them to remain, again repeating the treatment at intervals of a day. If necrosis of bone be present, it is good practice to alternate the iodine treatment with aromatic sulphuric acid. Cases generally yield to this treatment in from six week to three months. I have seen cases not yielding to treatment for nine months. There are other and shorter methods in the treatment which perhaps some would prefer—the cutting down through the body of the tumor, by making a crucial incision and scraping out the contents of the sac, afterwards allowing nature to do the rest—but I do not believe it is the best or safest way. There is surely a much greater loss of structure, which is never restored as in the former method by

granulation, after the secreting cells have been destroyed by medicinal applications of iodine and sulphuric acid treatment.—*Medical and Surgical Reporter.*

ARTICLE V.

THOROUGHNESS.

BY L. P. DOTTERER, D. D. S.

[Read before the South Carolina Dental Association.]

Though scarcely more than a novice in the vast field of Operative Dentistry, I have gleaned sufficient experience from observation and practice to know that THOROUGHNESS is the surest means of success.

Just as the tillers of the soil sow their seeds, watch their crops, and reap their harvests, so must we do our duty, advise our patients as to the best means of preservation, and would that I could say, reap our harvest. There has been so much written upon this subject that I have nothing *new* to say, but will touch upon several points, and in giving my idea of thoroughness, as there applied, I may draw out some discussion.

The first step towards the preparation of the mouth for dental operations is the removal of calculus and decayed fangs. Let this be done in a manner that will *insure future cleanliness*, where the proper after attention is given on the part of the patient.

As regards the preparation and filling of cavities, there are so many conflicting conditions, that we must be governed entirely by the case before us; but to be thorough in our preparation, we must so shape the cavity as to have the walls nearly plumb, uniform margin, slightly undercut. In proximal cavities there may be a groove or pit at cervical wall, but do not have it too near the margin, on account of its liability to produce fracture, and consequent failure at that point. On grinding surfaces, cut out all fissures leading into cavity, and be careful to have no angles.

The margin, after all, is the most important point; for just here failure begins, especially at the cervical wall, and care should be taken to thoroughly remove all softened structure, and aim to reach a solid

foundation. These margins should be carefully trimmed and burnished, and thus our cavity is ready for the filling.

We often hear practitioners decry the rubber dam, and boast of their skillful use of the napkin; but, gentlemen, many are the failures consequent! For in deep proximal cavities, the dam is invaluable in keeping guard against oozing moisture from the gums, which, without this precaution, will flow upon the filling without our knowledge. The dam adjusted, we proceed to form a mass of non-cohesive gold, and where the walls are strong enough, we can continue with this material throughout. But where cohesive gold is necessary, we should cover our borders, as far as possible, with soft foil; for this is more adaptable to the walls. Another advantage to be found in non-cohesive gold, is its pliability, ease of starting, and rapidity in finishing. We should thoroughly condense from beginning to end, whatever may be the kind of foil used.

Filing and finishing is too often hurried through, leaving a surplus of material at the cervical wall, or lapping the edges—another sure cause of failure; and every care should be directed to finish in such way that an instrument passing over the line of demarkation cannot detect it. After filing, we would use pumice, either on a strip of orange-wood, or by some other convenient means, and then polish. The same general rule holds good in amalgam work, and the main cause of failure in these cases is that lack of thoroughness in finishing.

In grinding surface cavities, where the enamel leading thereto is funnel-shaped, we often introduce too much amalgam, extending it beyond the margins of the cavity, and finishing to a fine edge. This material, when hard and bit upon, will fracture perpendicularly around the margins, giving the finishing a bulged appearance, and exposing a V-shaped crack, which will invite decay. Consequently, we should remove all surplus material, and finish at the very margin of the cavity. When gold is used, this precaution is not so necessary, as the edges of a gold filling will not fracture. Since we do not have to mallet amalgam, it is natural to suppose we don't require firm margins, but this is a mistake; and as much, or even more care should be exercised in the preparation of a cavity for amalgam than gold, as tooth-structure seems to waste away more rapidly from the former.

Let our motto be, "Whatever is worth doing at all is worth doing well." If applying arsenic or a disinfectant, cover it with gutta-percha, for the patient may be delayed a few days longer than we anticipate; and what is worse than removing a foul piece of cotton, and finding the tooth in a poorer condition than we left it? If we introduce a temporary stopping on account of exposure or frailty, let it be done thoroughly; and after relating its importance to the patient, caution her to return at a certain time for its removal and permanent filling.

We must be teachers at our chairs, if we wish the public to appreciate us, and we should instruct patients in the proper care of their teeth by an intelligent and thorough use of the brush, pick, etc.

Such is the importance of thoroughness in dental operations. This paper does not half express it, but for fear of trespassing too much on your valuable time, I commend these ideas to your criticism.—*Southern Dental Journal*.

ARTICLE VI.

WHAT FILLINGS SHOULD WE USE?

DR. W. G. A. BONWILL, PHILADELPHIA.

When I look back at my commencement and reflect that my early practice was founded on what the older men in authority had published and taught, and how I feared to do other than they demanded, I shudder at the many teeth I extracted I now know might have been saved, with even the amalgam of that day. And I tremble at the advice *now* given by the authorities that *gold* only should be used as a permanent filling. Young men knew no better, but the older do. God forgive them, I cannot. While I do not belong to the disciples of the new departure, *so far as their theory is concerned*, I stand side by side with any person *who can save teeth by plastic materials*, where gold cannot be used. Better do this than persist with gold indiscriminately, and lose teeth, rather than stoop to conquer with *any article* that is *not gold*. The public are demoralized on the subject of *gold*. "Are you not going to fill my teeth with gold?" says nearly every new customer; "Dr. ——— would not think of using anything else." A city operator must have more than the usual quota of courage to stand before the societies and state "he has been using *amalgam* more freely of late." For the first eight years of my practice I would not touch it, because Doctors Elisha Townsend and J. D. White passed their anathemas on everything but gold and tin. I worked myself nearly to death with tin to find it preserves from caries but not from attrition. Since 1862, I have been feeling my way, and while I think I have reared many beautiful and substantial monuments of gold, and have perfected machinery with which to do it, yet I consume more amalgam than ever before.

A gold filling *properly* impacted, with cavity judiciously prepared, and the walls shaped as to forbid future decay, *will save*, irrespective of the frailty of their bony structure? But as thousands of teeth *cannot be so prepared*, both of strong and of frail organizations, and the circumstances

cannot be controlled, we should resort to something that will enable us the more surely to meet the issue.

To enumerate the many cases of peculiar character that forbid the use of gold, would be too great a task. Physical impossibilities lie in the way of every undertaking; and it is for the successful engineer, who is well acquainted with his material, and their relative strength and *adaptability* for his purposes, to so use each, that his design will be consummated, and which shall not by future wear, prove a failure. There is a fitness in every material that experience has proven to be specially adapted for a given work, and when this general law is recognized and we become first-class engineers, we shall the better see where we can adapt our materials to the work to be done, and we can be the more certain of success, for it is founded on the logic of mechanics and physical law.

Where is the dentist that first lays out his design and orders materials best adapted for specific portions of it?

As well say everything should be made only of iron, or steel, or wood, as that every tooth should be filled with gold; or, as *equally ridiculous*, that the amalgam or some one of the plastic fillings should be the only material used.

It is not *necessary* to found a *creed or departure* on a law of *incompatibility* to tooth substance. We need not look so far into the unknown and unknowable. We poor, short-sighted creatures must have the tangible; not a hypothesis on a *supposed theory*. Any one with half an eye can see just where the incompatibility is; not between gold and dentos, but between dentos and untutored and unskilled brain and hands to *carry out the law of adaptibility*—the correlation of forces involved.

One skilled in the use of the mallet, with the rubber-dam and a substantial starting point, with walls ever so frail, can perfectly impact and complete the work in gold filling, *provided the surroundings are there*. But allow *one little vacuum* between the tooth substance and the filling, and a *capillary tube* will be formed to suck up *fermentable material*; and the *acid generated* will act on the tooth whether it be filled with gold, amalgam, oxyphosphate, or gutta percha. A thousand capillary tubes making porosity in the gold or the amalgam, will not do it; but if there is one, however small,

between dentos and filling, destruction is sure.—*Transactions of the Odontological Society of Pennsylvania.*

ARTICLE VII.

SOME METHODS OF SEPARATING TEETH WITH WEDGES.

BY DR. DWIGHT M. CLAPP, OF BOSTON.

[Read at the joint meeting of the Massachusetts and Connecticut Valley
Dental Societies, held at Worcester, Mass., June, 1885.]

Among the many disagreeable and annoying, not to say painful, things that patients have to suffer at the hands of dentists, nothing, perhaps, is received with greater dread and disgust than the announcement that the teeth must be "wedged" before filling. Some, a small minority among us, I think, always fill without previous separation. In regard to the necessity for it, I will enter no argument here, but only say that personally I am a firm believer in wide spaces between the teeth at their necks, and labor to the best of my ability to obtain this result. It is most likely that many of you are using the same means that I am to get the desired room for filling, but by presenting and discussing the subject, it is possible we may obtain some help in doing what I fear the most of us find, at times, difficult and perplexing. For a long time rubber was about the only thing used for separating. It has some good qualities and many bad ones. It probably causes more pain and annoyance to the patient than any other wedge. Its liability to slide into contact with the gum, causing great pain and soreness, and even suppuration, has caused me to entirely abandon its use, I am willing to admit that it may be used successfully sometimes. The best rubber to use, if it must be used at all, is that of which the most inelastic tubing is made, or the erasers sold by stationers, cut into suitable shape. Wedges of wood are well adapted to cases where the sides of teeth to be wedged are nearly parallel, or where there is less space at the gum than at the points of the teeth. The wedge should be about as wide as the length of the crown, that is, it should extend from the cutting edge to the gum, nearly. It should be so shaped and trimmed as to not irritate the tongue or cheek.

One advantage of the wooden wedge is that it is more cleanly than tape, cotton, or silk. This same class of teeth, those with nearly parallel sides, can be separated as successfully, and I think with less pain, with tape. Linen tape of various widths and well waxed is the best. It should be folded so as to be of proper width and thickness, and then drawn into place. A sharp knife is preferable to scissors for cutting off the ends. The tape should be thoroughly waxed, which assists materially in getting it between the teeth, and renders it more cleanly when left in the mouth for several days. In teeth with cavities so situated that cotton can be crowded in with sufficient force, this is one of the best wedges that can be used, as regards both effectiveness and comfort. It is necessary to so place the cotton that the force of expansion will be exerted against adjoining teeth and not expanded within the cavity. By once changing the cotton, space enough can generally be obtained. It is difficult to adjust and keep wedges in place between teeth having more or less space at the gum, and touching only at a small point near the cutting ends. It is in these cases that ligatures of various kinds serve an admirable purpose. Take for instance, the superior central incisors. These usually have but a small point of contact, with considerable space between them at the gum, and it is very difficult to put in a wedge of rubber, wood, or tape, that will not slip up against the gum, or come out altogether. If a ligature is used, the knots can be so tied that the string will clasp the point of contact in such a manner as to hold it quite firmly in place. There are many ways of making the knots; one is to pass the silk once between the teeth, then tie a surgeon's knot; but, before drawing it up, pass one of the ends again between the teeth, and then draw the knot so it will wedge from the gum towards the cutting ends; draw it closely, then finish by tying so that the last knot will be at the labial, or palatal side of the teeth. Another way is to make a series of knots like a chain stitch in crochet work, thus enlarging the silk for a suitable length; draw this between the teeth and tie as before, omitting the first knot that is drawn between the teeth. Another, and a very good way of enlarging the ligature, is, after well waxing it, to roll a little cotton around the silk as you would around a broach for wiping out a root canal, and draw this between the teeth and tie the same as when the silk is knotted. Still another method, easy of application and very effective in almost all cases where there is a cavity in one or both of the teeth, is to secure a pellet of cotton with the ligature. The silk is placed between the teeth in some of the before-mentioned ways; a pellet of cotton

is forced into the cavity, projecting against the adjoining tooth, then the silk is tied firmly around the cotton. The swelling of the cotton and silk will make all the space necessary between any of the front teeth with but one application. The bulging of the cotton into the cavity or cavities, caused by tying the silk around it will hold it securely in place. This makes by far the most satisfactory wedge I have ever used, and, so far as I am aware, is original with me. It is sometimes well to open the cavity slightly with an excavator or chisel before wedging, so that the cotton will be more readily retained. For bicuspid and molars more than one application may be needed if much space is required. Quick wedging is sometimes possible, and when it can be done readily is usually desirable. Teeth that move easily may be separated sufficiently for operations by placing a wedge at the point of contact, and another near the gum, applying force gently with the hand, or light blows with a mallet, first on one, and then on the other, until wedged enough. Then remove the wedge that interferes most with the operation, leaving the other in place. Another way that often works well with children and with teeth that move readily, is to insert a large piece of rubber and let it remain from fifteen to twenty minutes, when the rubber will have opened a considerable space. A wooden wedge will keep the teeth from springing together while the work is being done. The appliances designed by Drs. Perry, Bogue and others, for making rapid separations, I have not used, but hear favorable reports in regard to them. Having spoken of rapid and semi-rapid separations, it is left only to speak of a method which works very slowly. It applies, as a rule, to the bicuspid and molars only. In many cases where there are large cavities between these teeth, and often, when it is desirable that they should be filled with what I think is very properly called a "treatment filling," it is well to fill the entire space between the teeth with gutta-percha. In the course of a few months the process of mastication will force the gutta-percha toward the gum, and on removing what has not worn away the teeth will be found well separated, the cervical margins well in view, and the cavities in good condition for a metal filling.—*Archives of Dentistry*.

ARTICLE VIII.

COCAINE.

WALTER W. ALLPORT, M. D., D. D. S., OF CHICAGO, ILL.

The introduction of cocaine as a local anæsthetic, and the more general use of peroxide of hydrogen (H_2O_2) in the treatment of dental and oral diseases, are the principal advance made in the medical department of this practice during the year for which this report is made.

The two forms of cocaine which have been most generally used in surgery are the hydrochlorate and the oleate.

In operations in the mouth, involving the mucous membranes, together with the immediately subjacent tissues, these preparations have proven so efficient there is little question of its value as a local anæsthetic in such cases. But its action on deeper structures, such as involve the roots of teeth, is so uncertain as to render its practicable benefits questionable in the operation of extraction. In the surgical treatment of pockets caused by pyorrhea alveolaris, the anæsthetic effect of this agent is often so great as to render this sometimes very painful operation comparatively painless, and its employment in such cases should rarely be dispensed with. In the treatment of hypersensitive dentine, as well as in the removal of tooth-pulps, its action as an anæsthetic has, under some circumstances, seemed to be all that could be desired. But in far the greater number of cases it has proved of little practical value. More recently, however, a new form of cocaine, known as the citrate, has been introduced in Germany by Merck, and is now being manufactured by McKesson & Robbins, of New York. In a series of experiments, conducted by Dr. John S. Marshall, of Chicago, it has been shown that for operations on sub-mucous tissues, or in the extraction of teeth, it seems to possess no special advantages over the preparations previously named. But when applied to dentine or the pulp, its action—though not always positive—seems to be more reliable, especially on the

dentine, and gives promise of better results. Under favorable conditions it produces anæsthesia of the parts in from five to ten minutes, and the duration of the effect is of sufficient length to afford time for the preparation of the cavity. This effect has, in some cases been prolonged for more than an hour. The pulp has been extirpated without pain after the drug has been applied in from three to twelve minutes.

If the citrate of cocaine be kept in solution for more than three or four days it decomposes and loses its active properties. As introduced by Mr. Merck for dental purposes, it is made into pills by incorporating it with gum tragacanth dissolved in glycerine, each pill containing 1/8 grain of the citrate. In this form it keeps well. A pill is applied to the sensitive cavity and covered with a cotton pledget, moistened in tepid water. It should be allowed to remain from five to twelve minutes, when—if at all—the desired result is produced. In twenty per cent. of the cases where this remedy has been employed it has proven unsuccessful, but it is hoped that this percentage will be reduced by a better knowledge of the drug and the improved methods of its preparation and use.

With this in view, and at the suggestion of Dr. Marshall, McKesson & Robbins are now manufacturing granules containing one-sixteenth of a grain of the citrate of cocaine, without glycerine or any other saccharine excipient, so that the obtundent may act more promptly than it can in the presence of sugar.—*Address at American Medical Association.*

Editorial, Etc.

UNIVERSITY OF MARYLAND, DENTAL DEPARTMENT.—The fourth Annual session of this institution opened with a much larger number of matriculates than ever before in its history, and the number is so rapidly increasing that the present class of seniors and juniors bids fair to be larger than any preceding one.

The reputation of this school has never been sullied by the graduation of students for fees irrespective of professional ability, and the consequence of such a course as has been steadily pursued since its organization, has been to give a professional standing to its diploma which that of no other dental school excels. The present class consists of representatives from all parts of this country, and also Germany, France, South America, Canada, and even Turkey. Many states of this country are largely represented, such as New York, Georgia, Virginia, Pennsylvania, South Carolina and Maryland especially, and also the New England states, while nearly every other state is represented.

Students who have passed a session at other dental schools have entered on a second session at the University of Maryland, Dental Department, to complete their course of study and receive its diploma, and not one of the hundreds of students who have attended a course in this institution, has ever gone elsewhere to graduate. In matriculating the present class, the resolutions adopted by the National Board of State Dental Examiners have been strictly adhered to, and many applicants of this country and Europe have been refused admission who desired to make their attendance obligatory on graduation after ONE session's attendance.

The Infirmary and Laboratory practice is not excelled in size if equalled by that of any other dental school, and the records will show hundreds of gold fillings credited to the individual practice of students for both the regular winter and summer sessions. No other school can offer greater facilities for practical instruction, nor present more complete equipment as to building and appliances than this Dental Department. Dental practitioners are cordially invited to visit the University and inspect the specimen work of its graduating classes deposited in the museum. Large and valuable contributions from all parts of this country and also from

Europe are almost daily being received for the Museum, which will compare favorably with that of any other dental school for valuable pathological specimens, which are also utilized for illustrating the lectures of each course.

CORRESPONDENCE.—The following letter was received from a prominent dental practitioner of Georgia for publication:

AUGUSTA, Sept. 30th, 1885.

EDITOR "AM. JOURNAL OF DENTAL SCIENCE":

DEAR SIR: According to my knowledge of the proceedings of the organizations known as the "National Board of Dental Examiners," and the "National Association of Dental Faculties," it was determined that no dental school would be regarded as reputable that did not after June, 1885, require TWO FULL SESSIONS OF FIVE MONTHS EACH IN SEPARATE YEARS FOR GRADUATION. The only exceptions made being those who after graduation in medicine had passed one year in the study and practice of clinical dentistry, and also those who had attended a previous session at a reputable dental school. I believe that the American Dental Association also adopted the same rule. Am I not correct? I therefore ask how it is that the dental school of Vanderbilt University is permitted to offer graduation at the close of but one session, to a student of this city who has passed one session only, and that very irregularly, at the Georgia Medical College? I also ask how the same school can offer similar inducements to #/ it has done, and yet be declared reputable? Was it for the purpose of permitting such violations of the rules adopted by the different organizations referred to, that the "National Association of Dental Faculties" allowed the dental school of Vanderbilt University to abstain from becoming a member of that Association for the present year, and accorded to its Dean the privileges of the floor at its late meeting in Chicago? I cannot see why some schools should be compelled to conform to a rule that others may violate with impunity, and I think that the State Boards of Dental Examiners of both my own state and South Carolina should investigate the matter and act accordingly.

Respectfully, &c.,
"JUSTICE."

We can only reply to the above letter by stating that several students who as we had learned from their preceptors, intended to matriculate in the Dental Department of the University of Maryland, on discovering that they would be required to attend two sessions in the institution, had, we are informed, been induced to go to Vanderbilt by the promise of graduation on one session's attendance.

EDITOR OF "AM. JOURNAL OF DENTAL SCIENCE."

Bibliographical.

A Series of Questions Pertaining to the Curriculum of the Dental Student.—Embracing Dental Histology, Dental Pathology, Dental Surgery, Dental Prosthesis, Dental Metallurgy, Dental Materia Medica and Therapeutics, Anatomy, Physiology and Chemistry. By Ferdinand J. S. Gorgas, A. M., M. D., D. D. S., University of Maryland. Publishers: W. K. Boyle & Son, Cor. Baltimore and St. Paul Streets, Baltimore, Md. 1885. Price, \$1.50.

This work comprises leading questions on all the branches belonging to the course of study pursued by the dental student, and its object is to facilitate the study of dental science and its collateral sciences.

Some years ago the author published a small work embracing in an abridged form, questions on Dental Science, and the favor with which it was received by the students of his class, was such as to rapidly exhaust the limited edition, and induce him to prepare, when opportunity offered, a more complete series of questions embracing the entire curriculum of the dental student, and even that of the medical student so far as related to Anatomy, Physiology and Chemistry.

The present work, therefore, is the result of such an intention, and is presented with the hope that it may accomplish the object for which it has been prepared.

Practical and Analytical Chemistry.—Being a Complete Course in Chemical Analysis. By Henry Trimble, Ph. G., Professor of Analytical Chemistry in the Philadelphia College of Pharmacy. Illustrated. Publishers: P. Blackiston, Son & Co., Philadelphia. 1885. Price, \$1.50.

The object of this excellent work is to place before the student of pharmacy and medicine a compact course of analytical chemistry.

The distinguished author believing that the study of Qualitative Analysis should be preceded by some practical experience such as relates to the preparation of the important gases and a few of the salts, devotes Part

First to the consideration of Hydrogen, Chlorine, Hydrochloric Acid, Oxygen, Nitrogen, Ammonia, Nitric Acid, Carbon Dioxide and the preparation of such salts as Potassium Chloride, Potassium and Sodium Tartrate, Ammonium Nitrate and Oxalate, Calcium Phosphate, Magnesium Sulphate, Carbonate and Oxide, Aluminium Hydrate, Ferrous Sulphate, Ferric Sulphate and Hydrate, Copper Sulphate and Lead Acetate. Part Second is devoted to Qualitative Analysis, and Part Third to Quantitative Analysis, together with a description of apparatus, and the processes of filtration, evaporation, crystallization, ignition, etc. The work extends over nearly one hundred pages, and is a valuable text-book for the student.

Chemical Problems.—By Dr. Karl Stammer. Translated from the second German edition, with explanations and answers, by W. S. Hoskinson, A. M., of Wittenberg College, Ohio. Publishers: P. Blackiston, Son & Co., Philadelphia. 1885. Price, 75 cents.

The text is in the form of questions to which answers are given at the end of the volume, which comprises one hundred and nine pages. Part First relates to the recognized elements, and Part Second to approximate ratios, temperature, atmospheric pressure and mixed problems, making a compact and useful text-book for the study of chemical problems.

Quiz Questions.—Course on Dental Pathological and Therapeutics, Philadelphia Dental College. Prof. J. Foster Flagg, D. D. S. Answered by William C. Foulks, D. D. S. Third edition, Revised and Enlarged. Publishers: The S. S. White Dental Manufacturing Company, Philadelphia, New York, Boston and Chicago.

This excellent series of questions and answers on Dental Pathology and Therapeutics is again offered to the dental profession as a work of reference in daily office practice, and is the only book that contains in a condensed and practical form the facts and principles of these subjects, as enunciated by Professor Flagg. Commencing with General Principles, these questions and answers relate to Deciduous Teeth, Pathological Dentition, Lancing, Permanent Teeth, Dental Caries, Sensitive Dentine, Galvanic Action, Pulp Protectors, Obtunding Applications, Pulp Capping,

Pathological Conditions of Pulp, Dental Exostosis, Malformed Teeth, Periodontitis and Alveolar Abscess, the whole constituting a work of great value to all engaged in the practice of dentistry. The work is interleaved with blank pages for notes, etc., and gotten up in a neat and excellent style. We take great pleasure in commending this treatise as a valuable adjunct to the regular text-books of the profession.

Monthly Summary.

PEROXIDE OF HYDROGEN.—*Dr. W. W. Allport, Chicago.*—The peroxide of hydrogen ($H_2 O_2$), though not a new remedy, has only within the last few years gained much prominence in the treatment of surgical diseases. One of its uses in dental and oral surgery is in blind or deep-seated abscesses, such as arise from roots of diseased teeth. As the tendency of pus is always downward, when these cases occur in the lower jaw it is not infrequent that the abscess, if left to itself, and sometimes even after the tooth is extracted, will point through the external tissues at the lower margin of the jaw, and occasionally downward between the muscles of the neck, and open at various points, even as low down as the clavicle. The usual treatment is to extract the tooth and evacuate the pus through the alveolus, but it often happens that the formation of pus and the continuance of suppuration is not checked, and the abscess points, or is opened through the external tissue of the face or neck, leaving, when healed, a disfiguring scar.

By injecting peroxide of hydrogen into such abscesses before they point through the external tissues, this serious disfigurement can usually be averted, and the suppurative process is materially shortened. It is also a valuable aid for the evacuation of the purulent contents of the antrum of Highmore, in catarrhal and suppurative inflammations, and especially where the sinuses are divided into two or more pockets by bony septi. These cases are often protracted by the inability of the surgeon to perfectly evacuate them. But with this preparation it becomes a simple matter after access has been gained to the cavity by the extraction of a tooth or the perforation of its external wall in the proper place at the juncture of the cheek with the alveolar border. A free opening must always be made for the escape of the contents, in order to avoid pressure from the rapid evolution of gas. Two or three applications of a dram each is usually sufficient to completely empty the sac.

It is used with the most gratifying results in the treatment of pyorrhea alveolaris, and is an invaluable agent in treating pulpless teeth, as by its action all decomposed matter from the pulp chamber and dentinal tubuli is readily ejected, thereby removing the most frequent cause of discoloring of this class of teeth, of inflammation of the peridontal membrane, as well as alveolar abscesses.

The efficacy of peroxide of hydrogen depends on the case with which it is decomposed into oxygen and water. Pus is one of the many substances which causes this decomposition. Hydrogen peroxide acts first chemically and then mechanically. When the decomposition takes place the oxygen is set free and escapes from a liquid to a gaseous form; this expansion of the gas distends the pus cavity, and as it escapes from the orifice, it carries much of the pus with it, and its application should be repeated till all purulent accumulations are evacuated. The liberated oxygen, being in a nascent or active condition, rapidly oxidizes the products of suppuration, and destroys many of the micro organisms of suppuration.^[A] Hence it is a disinfectant and anti-septic.

Finally, peroxide of hydrogen, after its decomposition, leaves no material in the system which is foreign to the system, and it is, therefore, one of the most efficient and harmless disinfectants and anti-septics that can be used, in all forms of purulent inflammation.—*Address at American Medical Association.*

ALCOHOLIC PARALYSIS.—The immediate and transient effects of an excessive quantity of alcohol upon the human nervous system, whether they are manifested in the form of drunkenness, or of delirium tremens, or of an acute attack of insanity, are well known. Scarcely less evident are the effects produced upon the nervous system by a less excessive, but a more prolonged, abuse of alcoholic drinks. These effects may be manifested either in a general failure of physical and mental power, or in a form of disease closely resembling progressive paralytic dementia, or in various forms of chronic insanity, or in epilepsy, or in neuralgia, or in paralysis. In the acute form of alcoholic poisoning, no change in the structure of the nervous system has been found, except that the meninges in common with the internal organs and the mucous membranes are the seat of a very decided injection and slight exudation. In the chronic form of alcoholism, a number of pathological changes have been discovered in the nervous system, which, however, vary greatly in different cases.

Of late years the paralysis which results from the abuse of alcohol has been accurately described by numerous observers, and the attempt has been made to discover the lesion of the nervous system which is associated with this form of paralysis. Two cases which are reported by Dr. Henry Hun, of

Albany, in the *American Journal of Medical Sciences* for April, 1885, are typical examples of this disease, and contribute to a better understanding of it.

Dr. Hun has collected the recorded cases of alcoholic paralysis, and from their study he holds that we are justified in regarding it as a special form of disease with the following symptoms: After a number of cerebral and gastric disturbances due to the alcoholic poisoning, the symptoms of the disease proper commence with neuralgic pains and paræsthesiæ in the legs, which gradually extend to the upper extremities, and which are accompanied at first by hyperæsthesia, later by anæsthesia, and in very severe cases by retardation of conduction of pain. Along with these symptoms appears a muscular weakness, which steadily increases to an extreme degree of paralysis, and is accompanied by rapid atrophy and by great sensitiveness of the muscles to pressure and passive motion. Both the sensory and motor disturbances are symmetrically distributed, and the paralysis attacks especially the extensor muscles. In addition to these motor and sensory symptoms, there is also a decided degree of ataxia. The tendon reflexes are abolished and vaso-motor symptoms, such as œdema, congestion, etc., are usually present. Symptoms of mental disturbance are always present in the form of loss of memory and in transient delirium.

The lesion is in all probability a degeneration of the peripheral nerve fibres and of the nerve cells in the cerebral cortex, together with a chronic congestion or inflammation of the pia mater. This lesion explains well the symptoms, although it is curious that alcohol should not attack the spinal cord, but only the highest and lowest part of the nervous system, if one may so call the cortex of the brain and the terminal branches of the peripheral nerves.—*Detroit Lancet*.

NOURISHING THE TISSUES OF THE TEETH.—Dr. Frank Abbott says: For a number of years past I have entertained the views that there was some difficulty existing not due to an insufficiency of lime-salts, which occasions so many faulty and imperfectly formed teeth. With that idea in view I have advised exercise in the open air and other kinds of treatment for some patients which would favorably affect their digestion. That the food ordinarily taken contains sufficient lime-salts to form and to sustain the teeth I have no doubt (except in cases of extreme anemia, during gestation

and lactation). I believe the fault is beyond that. There is a lack of proper nourishment of the tissues, due to imperfect digestion, which depends again on the proper "nerve tone." I believe this to be the real cause of the difficulty. It is reasonable to suppose that any *tonic*, whether taken in the form of exercise in the open air, or in any other form, which affects favorably other portions of the body, will affect the teeth favorably as well. When we have *ascertained the functions of the great nerve-centers*, and those functions are assured, then we will probably have more perfectly formed teeth.—*Items of Interest*.

Footnotes

[A] See Gradle on "Bacteria and the Germ Theory of Disease," pp. 39 and 151.

*** END OF THE PROJECT GUTENBERG EBOOK THE AMERICAN
JOURNAL OF DENTAL SCIENCE, VOL. XIX. NO. 6. OCT. 1885 ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.